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DAILYKHATA

Deliverable 2: Problem Identification, Stakeholder Identification & Requirement Elicitation

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1. Introduction

This deliverable explains how the DailyKhata expense-recording problem was identified, which stakeholders are affected, and how the reported findings inform requirement candidates. It develops the user analysis in Deliverable 1 without repeating the full product description or persona profiles. DailyKhata remains a personal expense management application for Pakistani users, with personal and household recording as its main purpose.

The analysis uses the interviews, informal observation and existing-system examination reported in the earlier project report. It also uses document analysis to connect those findings with the agreed project scope. Reported fieldwork and project design decisions are distinguished throughout: a feature can be justified by the proposed solution without having been explicitly requested in an interview. [1, 2]

The requirement candidates provide the link to Deliverable 3. They describe the behaviours suggested by the evidence and project decisions; D3 formalises them as business, functional and non-functional requirements. This approach follows the assignment's emphasis on project-specific evidence and the progression from stakeholder needs to system requirements. [3]

2. Problem Identification

2.1 Existing Workflow

The reported workflow begins when a person makes a purchase and either keeps the receipt, writes a note or relies on memory. An expense may be entered later in a diary, phone note or application. Later entry requires the person to reconstruct the date, amount and purpose. At the end of a week or month, separate entries must be combined to understand spending. [1, pp. 2-3]

Each method preserves only part of the information needed for useful review. A receipt contains detail but may be lost or difficult to read. A note can be convenient but may lack consistent categories. Memory can preserve the general purpose of a purchase while losing its precise amount. These limitations explain why capture and later interpretation need to be considered together.

Existing method	Limitation in the reported workflow	Implication for DailyKhata
Receipts and diaries	Information is scattered; finding purchases and adding totals takes manual effort.	Digitise receipt information into reviewable, searchable records.
Phone notes	Free text needs interpretation and manual aggregation.	Convert entries into consistent fields and categories.

Existing method	Limitation in the reported workflow	Implication for DailyKhata
Memory	Small purchases may be omitted or recalled imprecisely.	Support quick cash entry, past dates and later correction.
Expense applications examined by the team	Repeated field entry and no item-price view were reported in the tested workflow.	Investigate shorter capture paths and useful product history.

The problem is therefore more specific than a general lack of financial awareness. Users need a manageable way to create a record and enough structure to retrieve and understand it later. An application that only displays charts would still depend on reliable capture, while capture without correction could produce misleading analysis.

2.2 Interviews

The earlier D2 reports approximately one week of short interviews and informal observation. It identifies three interviewees: Ayesha Tariq, Hamza Sheikh and Usman Raza. Conversations reportedly lasted 10 to 15 minutes and took place in person or by phone. The initial questions explored current recording methods, update frequency, abandoned tracking attempts and conditions that might encourage continued use. [1, p. 3]

Interviews were appropriate for exploring why participants found recording difficult and which input methods they preferred. The available report contains attributed statements and summarised findings, rather than transcripts or recordings. The following quotations are reproduced from that report and have not been independently checked against primary interview records.

Ayesha Tariq: "If I could just take a picture and it sorts itself out, I would actually keep doing it. Typing it all in is why I stop after two weeks."

Hamza Sheikh: "I deleted the last app I tried because filling in every field felt like an assignment, not something I would do on my own."

Usman Raza: "Even if I could just say the amount out loud when a customer pays, that alone would stop me from losing track."

Ayesha's statement links continued recording with reduced typing. Hamza describes repeated form completion as burdensome. Usman's statement suggests interest in spoken capture, but its customer-payment example concerns income. It supports the general theme of capture effort, not a sales ledger or payment-processing requirement. Voice-based expense entry remains Should Have.

The report identifies Ayesha as 34 and Hamza as 21. It gives Usman's interview age as 46, whereas the agreed design persona is 41. The persona remains 41, and the participant age requires confirmation. Bilal Ahmed and Zubair Hassan are design personas, not additional reported interviewees.

2.3 Observation

The reported informal observation involved family members, roommates and classmates in home kitchens, hostel rooms, the university canteen and a neighbourhood shop. The team watched whether people wrote down small purchases, opened an application, retained a receipt or took no recording action. The source does not provide an event count or a participant list. [1, pp. 3, 10-11]

The observation narrative reports that entries were often delayed or omitted and that later recall could be rounded or incomplete. It also reports that phone notes did not provide immediate category totals in the observed workflow. These findings complement the interview accounts by considering what happened near the purchase itself.

The distinction between action and interpretation matters. Seeing no entry at a purchase does not prove that the person never recorded it later. The design implication is to support both quick entry and retrospective correction, rather than to claim a measured omission rate. The supplied Design Thinking material supports this separation of observed behaviour, explanation and inferred need. [4]

2.4 Existing System Analysis

The earlier D2 reports that each team member spent 30 to 45 minutes testing Cashew, Wallet and an unnamed bank application on Android. The task was to add a grocery-style expense and look for history for repeated items. The reported findings were repeated typing, limited adaptation of local labels in that test, and no item-price view in the examined workflow. [1, pp. 10-12]

This examination provided a practical comparison for entry effort and information retrieval. It helped the team identify questions for DailyKhata: whether local category labels would be understandable, whether repeated purchases could retain item detail, and whether capture could reduce manual transcription.

The report does not supply application versions, test dates, account tiers, screenshots or a detailed procedure. Its findings are therefore a historical account of the team's examination, not a statement about current product capabilities. Mint and Notion-based sheets mentioned in earlier market discussion were comparison context, not additional tested systems. No claim of universal competitor limitations is needed to justify the proposed DailyKhata workflow.

2.5 Key Findings

The evidence can be summarised without assigning a separate identifier to every statement. The table also shows which conclusions come from project analysis rather than fieldwork.

Evidence basis	Main finding	Requirement implication
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Evidence basis	Main finding	Requirement implication
Ayesha and Hamza interview statements	Repeated typing and long forms discourage recording.	Receipt capture, short manual entry and text-based cash entry.
Usman interview statement	Spoken capture is attractive in the reported example.	Consider voice expense entry as an optional route.
Requirement-focused interview questions	Participants wanted convenient input and confirmation of proposed categories.	Review and correction before saving.
Informal observation	Recording could be delayed, omitted or reconstructed later.	Retrospective dates, editable records and useful summaries.
Historical application examination	The tested workflow involved repeated entry and did not provide the desired item-price view.	Local labels, optional custom labels and product price history.
Project document analysis	The solution includes authentication, accounting structure, semantic products and a specialised assistant.	Preserve these design decisions when formalising requirements.

Three themes connect the methods: low recording effort, dependable correction and useful retrieval. Convenience alone is insufficient if a saved record contains the wrong amount or category. Equally, accurate records provide limited benefit if the user cannot find them or understand their totals. The proposed solution addresses this sequence from capture to review to understanding.

2.6 Problem Statement

Paper receipts and unrecorded cash purchases leave personal and household expense histories incomplete. Manual upkeep makes consistent recording difficult. Users need low-effort capture and reliable ways to review and understand spending.

3. Stakeholder Identification

The stakeholder analysis follows the assignment's required progression from industry to stakeholders. It uses the D1 personas as representative profiles, while distinguishing direct application users from people affected by its outputs. [2, 3]

Stage	DailyKhata application
Industry	Financial technology, specifically personal expense management.
Market	Expense-management applications for Pakistani users.

Stage	DailyKhata application
Market segment	Adults managing personal or household expenses with paper receipts and cash purchases. The indicative 18 to 50 age range is not an eligibility limit.
User profile	Smartphone users with varied digital confidence who need convenient capture, familiar categories, privacy and understandable spending records.
Persona	Ayesha Tariq, 34; Bilal Ahmed, 27; Hamza Sheikh, 21; Zubair Hassan, age not recorded; Usman Raza, 41. Their roles, goals and behaviours are defined in D1.
Stakeholders	Primary account holders, secondary shopkeepers and information recipients, the development team, supervisor, evaluators and conditional technical providers.

3.1 Primary Stakeholders

Homemakers and household budget managers, university students and salaried professionals are the primary users. Their interests guide the core recording and review workflow. They need the application to accept different purchase types while presenting clear values, categories and summaries.

The D1 profiles provide complementary perspectives. Ayesha emphasises household receipts and grocery prices; Hamza emphasises frequent small cash purchases; Bilal emphasises periodic monthly review. These are design priorities, not assumptions that every person in a group has the same habits. Later evaluation should include the three groups and allow for differences in digital confidence.

3.2 Secondary Stakeholders

Small shopkeepers are secondary direct users for petty cash and supplier expenses. Their needs can be served through the same expense ledger, receipt capture and summaries. They do not change the primary market or require sales, inventory, payroll or point-of-sale functionality.

Family members, accountants and financial advisers are secondary information recipients. Zubair represents a family member reviewing household spending with the account holder. Their interest is understandable output that the user chooses to share. This does not establish shared household accounts, collaborative editing, an adviser portal or guaranteed professional involvement.

3.3 Internal and External Stakeholders

The internal project stakeholders are the four-student development team, supervisor Nouman Ali and academic evaluators. The team is responsible for analysing needs, implementing the agreed functions and collecting evaluation evidence. The supervisor

and evaluators assess the relevance, clarity, feasibility and academic quality of the work. Planned review is not evidence that endorsement has already occurred.

OCR, AI and hosting providers become external technical stakeholders if selected. Their interfaces, costs, service availability and data handling affect feasibility. No provider is committed at this stage. Bank and payment partners are future stakeholders only because transaction integration is not part of current delivery. Privacy and regulatory considerations constrain design, but no regulator interview or legal assessment is reported.

3.4 Stakeholder Matrix

Stakeholder and role	Interest and needs	Impact on the proposed system
Primary account holders: household managers, students and salaried professionals	Low-effort capture, trustworthy corrections, privacy and useful spending review.	High influence on core workflow, terminology and usability evaluation.
Small shopkeepers: secondary direct users	Dependable petty-cash and supplier-expense records.	Inform expense scenarios without adding business-management functions.
Family members: secondary beneficiaries	Clear household totals shared by the account holder.	Inform summary readability; do not create household permissions.
Accountants and advisers: optional recipients	Understandable records and, if implemented, exported information.	Inform report clarity without requiring privileged application access.
Four-student development team	Feasible implementation, maintainability, affordable services and testable requirements.	High influence on design, integration and evaluation.
Supervisor and academic evaluators	Evidence-based work, clear scope and defensible validation.	High influence on academic review and acceptance of proposed targets.
Selected OCR, AI and hosting providers	Compatible service use and permitted data processing.	Affect latency, cost, reliability and exposure of submitted data.
Future bank or payment partners	Possible later transaction importing or integration.	No current implementation dependency.

4. Requirement Elicitation

4.1 Interview Findings

The interview technique was used to explore motivations that observation alone could not explain. Questions about abandoned notebooks or applications sought reasons for

stopping, while questions about preferred input methods explored acceptable alternatives. The reported 10 to 15 minute conversations involved Ayesha, Hamza and Usman. [1, pp. 3, 10]

The initial questions asked how expenses were recorded, how often records were updated, whether an earlier method had been abandoned and why, and what would encourage continued use. Three additional questions asked whether logging in under 10 seconds would help, whether typing, speaking or photographing was preferred, and what would make automatic categorisation trustworthy. The source does not establish a separate date or visit schedule for those additional questions.

The team interprets these findings as support for alternative capture routes and visible confirmation. A photograph can reduce receipt transcription; text and a short form can support no-receipt purchases. Category suggestions should remain editable before saving. Interest in speech supports its consideration, while its Should Have priority allows the minimum system to work without speech recognition.

The 10-second question indicates interest in low effort, not an accepted performance result. Image capture, network processing and human correction vary between tasks. D3 therefore separates measurable response behaviour from the user's total task time instead of promising that every purchase can be recorded within the interview prompt's suggested time.

4.2 Observation Findings

Observation was selected to examine actions around purchases rather than relying entirely on recalled habits. The reported settings and actions are described in Section 2.3. Delayed recording suggests that the application should allow a purchase date different from the creation date. Incomplete recall suggests that users need to revisit and correct saved information.

The lack of immediate totals in the reported note-taking workflow supports category totals and monthly summaries. This is a progression from an observed activity to a proposed behaviour: scattered entries create additional review work, so the application should aggregate the saved records consistently. It does not establish how often this problem occurs across the wider population.

For later validation, the important question is whether the proposed workflow supports the activity in practice. Users should be able to enter a past cash expense, locate it and recognise which month includes it. These are proposed evaluation tasks, not completed observations or user-test results.

4.3 Existing System Findings

Existing-system analysis was selected to examine concrete entry and retrieval workflows. The reported Android exercise involved adding a grocery-style expense and seeking repeated-item history. Its findings informed locally understandable labels, optional label

customisation and item-level price history, while the interviews provided a complementary explanation of entry effort.

Document analysis supplied the remaining design foundations. A full internal Chart of Accounts, stable category mappings and accounting-compatible records support coherent expense storage. Semantic product classification supports comparisons without merging brands or pack sizes. Authentication, saved-record maintenance and the specialised AI Financial Assistant complete the agreed solution. These are project decisions, not additional claims about what the three interviewees requested.

This distinction keeps the reasoning transparent. Fieldwork supports the need for convenient and trustworthy recording. The project design specifies how a coherent application will serve that need and provide later analysis. Formal requirements must preserve both sources without presenting technical design choices as direct quotations from users.

4.4 Requirement Candidates

The candidate identifiers below retain a simple link to D3. Each candidate groups a logical outcome of elicitation or document analysis. The priority column controls delivery commitment even when a candidate contains several related behaviours.

Candidate	Basis and proposed behaviour	Priority
DR-01	Receipt-entry effort and review preferences: photograph or upload receipts, extract structured information and allow correction before saving.	Must Have
DR-02	Long forms and delayed cash recording: provide short manual and text-based entry, with retrospective expense dates.	Must Have
DR-03	Interest in spoken capture: provide voice expense entry through editable text and review.	Should Have
DR-04	Trust in categorisation and local context: propose transaction and line-item categories with familiar labels, then require confirmation or correction.	Must Have
DR-05	Accounting design: maintain a full internal Chart of Accounts, stable category-to-account mappings and accounting-compatible expense records.	Must Have
DR-06	Fragmented records and difficult review: provide ledger history, search, filtering, category totals, monthly summaries and a dashboard.	Must Have
DR-07	Product-history needs and design: retain item-level prices and semantic product classifications without discarding original product identity.	Must Have

Candidate	Basis and proposed behaviour	Priority
DR-08	D1 solution and assistant design: answer spending questions and provide conversational analysis grounded in the user's records.	Must Have
DR-09	Privacy and record maintenance: authenticate users, support basic profile/settings, protect data and allow expense editing and deletion.	Must Have
DR-10	Presentation and sharing: offer English and Urdu localisation, CSV and/or PDF export, and custom labels for existing categories.	Should Have
DR-11	Optional review aids: offer simple visualisations, dedicated month comparisons and receipt image archiving.	Should Have
DR-12	Optional recording support: consider user-created categories, offline drafts, reminders and exportable household summaries.	Could Have

The candidates are intentionally broader than individual interface actions. D3 can combine related actions into testable requirements while retaining all included capabilities. For example, structured extraction can cover merchant, date, item, quantity and price fields together, provided review and missing-data handling remain explicit.

4.5 Quality Expectations

Quality expectations concern how well the functions support the intended workflow. They follow from D1's user needs, the reported friction in D2 and the requirements of handling personal financial records.

Quality area	Expected behaviour	Evaluation needed
Usability and responsiveness	Reduce repeated work; show processing, correction and recovery clearly.	Observe core task completion and measure ordinary interaction separately from OCR and assistant latency.
Security and privacy	Protect credentials, stored records and user-specific access, including assistant retrieval.	Test two-account isolation, safe uploads, communication and service data handling.
Device and network behaviour	Support ordinary Android-class devices and explain unavailable services.	Record device/network conditions and exercise failure and retry paths.
Data integrity	Keep totals, categories, account mappings and price history consistent after changes.	Test mixed receipts, rounding, edits, deletion and balanced journals wherever used.

Quality area	Expected behaviour	Evaluation needed
Assistant reliability	Ground claims in authorised records and acknowledge uncertainty or missing data.	Compare answers with known calculations and test ambiguous or unsupported requests.

These are evaluation needs, not measured achievements. D3 retains proposed test conditions and targets for review before implementation testing. Offline draft entry remains Could Have; reasonable network recovery does not require offline OCR or an offline language model.

5. Scope Summary

D1 establishes the full high-level scope. The candidate table above preserves its delivery priorities without repeating the complete feature catalogue. Must Have features define the minimum successful FYP; Should Have features are intended but deferrable; Could Have features are optional if time permits. Future Scope is potential later work, and Out of Scope is explicitly excluded from current implementation.

Three boundaries are particularly important when interpreting the evidence. First, household recording belongs to one account holder; family review does not create shared accounts. Second, shopkeeper participation supports expense capture only. Third, familiar category labels map to a structured internal Chart of Accounts, but users do not operate a full accounting system. Double-entry journals must balance wherever used; selecting double-entry remains an implementation decision.

The AI Financial Assistant remains a core, read-only capability for natural-language spending questions, summaries, period and product comparisons, recurring spending analysis, historical prices, unusually high recorded spending and personalised saving suggestions from recorded history. It does not provide investment, banking or tax advice, unrelated financial education, or unrestricted financial conversation. It cannot edit expenses or execute payments.

English and Urdu localisation remains Should Have, as do voice entry and general export. Local labels remain core. Dedicated household exports are Could Have. The exclusions and future opportunities defined in D1 continue unchanged, with D3 providing their detailed treatment. In particular, bank and wallet integration, shared household collaboration and complete business accounting are not introduced by the research findings. Historical pattern analysis does not commit the team to automatic recurring transactions, predictive anomaly detection or advanced recommendation systems.

6. Limitations

The available evidence is the earlier report's account of its research. Raw interview transcripts, audio, dated notes, consent records and observation logs are unavailable. There are no observation counts or completed questionnaires. The record contains no

reported survey, focus group, prototype trial or benchmark study. It therefore supports qualitative design reasoning rather than population-wide or measured-performance claims.

The team should confirm the interview context and Usman's actual participant age. Competitor versions, test dates, account tiers, the bank application's identity and test artefacts also remain unknown. A future examination could address these gaps, but no such work is claimed here.

The five personas should not be treated as a sample size. They represent design needs, with only three corresponding names appearing in the reported interview record. Likewise, proposed D3 evaluation participants and test receipts are future validation resources, not additions to this research. Student verification and supervisor review are still needed before these drafts become submission documents.

7. Conclusion

The reported research identifies repetitive entry, delayed cash recording and difficulty turning scattered purchases into understandable information. Stakeholder analysis shows that these needs can be addressed within one personal and household expense workflow. The requirement candidates connect those findings with the agreed accounting, product-history and assistant capabilities, providing a clear foundation for the formal specification in Deliverable 3.